

Indian Institute of Information Technology Kottayam

Department of Electronics and
Communication Engineering

and

Department of Computer Science
and Engineering
organizes

Six Day Online Faculty
Development Program

on

AI and Emerging
Technologies for Sustainable
Energy and Climate
Adaptation

(Online Mode)
25-30 November, 2024



Indian Institute of
Information Technology
Kottayam

Organizing Committee

Patrons

Dr. Prasad Krishna, Director, IIITK.
Dr. M. Radhakrishnan, Registrar, IIITK.
Dr. P. Mohanan, PIC Academics, IIITK.

Convenor

Dr. A. Ananth,
HOD, ECE, IIIT Kottayam

Dr. Rubell Marion Lincy
HOD, CSE-II, IIIT Kottayam

Coordinators

Dr. A Ravi, Asst. Professor, IIITK
☎ +91-8688535366
✉ draravi@iiitkottayam.ac.in
Dr. Della Thomas, Asst. Professor, IIITK
☎ 9444114708
✉ della@iiitkottayam.ac.in

Registration

Target Audience - Faculty, PhD Scholars, UG/
PG students, Industry personnel

(Seats will be assigned on a first-come, first
served basis)

Link for Registration

<https://atalacademy.aicte-india.org/signup>

Last date for

Registration: 24-11-2024

Free Registration



Indian Institute of Information
Technology Kottayam, Valavoor P.O.,
Pala, Kottayam- 686635, Kerala.

About IIIT Kottayam

Indian Institute of Information Technology Kottayam at Valavoor, Pala, Kerala is one among the IIITs that have been established as "Institution of National Importance" by MoE, Govt. of India under the ambit of IIIT PPP act 2017. IIIT Kottayam campus is situated at Valavoor, Pala, Kottayam - a picturesque location, spread over in 53 acres of land - allotted by Govt. of Kerala in the southern part of the Country commonly referred to as God's Own Country. B.Tech students are admitted through the JoSAA/CSAB counselling against the JEE main rank list. Institute offers 4 year B.Tech and B.Tech (Hons) programs in ECE, CSE, CSE with specialization in AI & Data Science and Cyber Security. In addition, Institute also offers Ph.D. programs in ECE, CSE and Computational Mathematics as well as M.Tech programme in (1) AI & Data Science (2) Cyber Security (3) Big Data & ML for the working professionals. Further, IIIT Kottayam offers Integrated M.Tech in AI & Data Science for working professionals.



Key Topics:

- AI in Sustainable Energy:
- AI applications in renewable energy sources (solar, wind, biomass, etc.).
- Smart grid management using AI.
- AI for energy forecasting and demand management.
- Emerging Technologies in Energy:
- Role of IoT in energy-efficient solutions.
- Blockchain for decentralized energy management.
- Quantum computing and its future role in energy.
- Climate Change and Adaptation:
- AI for climate modeling and predictive analytics.
- Big data and machine learning for climate resilience.
- Technologies for monitoring and managing carbon emissions.
- Case Studies and Applications:
- Smart cities and energy management systems.
- AI-driven environmental monitoring tools.
- AI-based innovations in transportation and smart grid technologies.

Hands-On Sessions:

- AI and ML algorithm implementation for energy efficiency.
- Hands-on with IoT devices for energy monitoring.
- Climate data analytics and visualization using machine learning tools.

Resource Persons

- 1.Dr. Anand Mukhopadhyay ,Mathworks
- 2.Prof. Nilesh J Vasa, IIT Madras
- 3.Dr.Kanendra Naidu, University of Tekonology Malaysia
4. Prof. Tiju Thomas, IIT Madras
- 5.Kasi Ponnappalli, GEOVIDYA.
- 6.Dr.Soumitra Satapathi, IIT Roorkee
- 7.Mr. S Bhavani Sankar, Effetronics Pvt. Ltd.
- 8.Prof. Jitendra Sangwai , IIT Madras
- 9.Dr. Ambika S, IIT Hyderabad
- 10.Dr. Ing. Parvathy A.S., Daimler Truck AG, Germany
- 11.Mr. Pavan Attavane, Advitiya R&D labs
- 12.Dr. Sharjad A J, COMSOL Inc.
13. Dr. V S S Kiran, CEO, Garudalytics

Experts from Academia and Industry from India and abroad will be delivering talks and hands-on sessions.

Potential Collaborations:

Renewable energy companies, research labs, and AI technology firms, Govt. & Intl. organizations working on climate adaptation strategies.

FDP on "AI and Emerging Technologies for Sustainable Energy and Climate Adaptation" would focus on equipping educators, researchers, and professionals with the knowledge and skills to integrate artificial intelligence and cutting-edge technologies into sustainable energy solutions and climate change adaptation strategies.

The main objectives of the FDP:

- To introduce participants to AI applications, IoT, and emerging technologies in the field of sustainable energy.
- To provide insights into how AI can be used to tackle climate change challenges and adaptation.
- To promote interdisciplinary research in energy systems and environmental sustainability.

Expected Outcomes:

Develop a deeper understanding of AI's role in energy efficiency and climate adaptation.

Foster research collaborations between academia and industry in sustainable energy.

Encourage the development of AI-driven projects for climate resilience.

